NANYANG TECHNOLOGICAL UNIVERSITY
PH4501/PAP710 MIDTERM EXAMINATION

Tuesday, Oct 6th 2021

Total: 100 points. Time allowed: 70 minutes

1. **Gaussian distribution (10 points)**

   Evaluate

$$\langle x^6 \rangle = \frac{\int_{-\infty}^{+\infty} dx\, x^6 e^{-\frac{1}{2}ax^2}}{\int_{-\infty}^{+\infty} dx\, e^{-\frac{1}{2}ax^2}}.$$

2. **Liquid-gas mixtures (10 points)**

   Consider the Gibbs potential plots in Fig. 1. Here x denotes the fraction of gas, increasing from left to right in the plot. Discuss briefly the phase the system is in as a function of x. Draw this diagram in the exam booklet and indicate on your plot where is the mixed phase region and how you construct it.

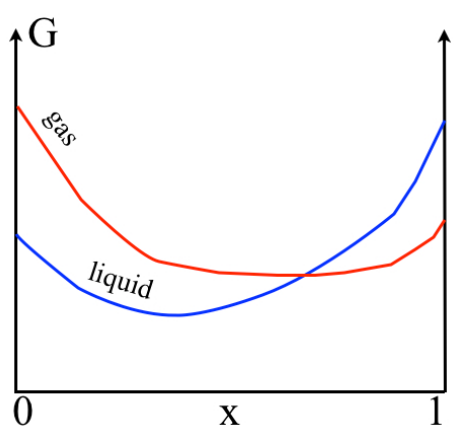


Figure 1: Gibbs potential plots for a simple liquid-gas transition in a binary mixture.

3. **Critical exponents (10 points)**

   Determine the critical exponent λ for the function

$$f(t) = At^{2/5}(t+B)^{1/4}$$

4. **Thermodynamic potentials (20 points)**

   (a) The equilibrium state of a system is a state of maximum entropy and that S is a concave function of its extensive variables, i.e.

$$\left(\frac{\partial^2 S}{\partial E^2}\right)_{V,N} \leq 0.$$

Explain why E is a convex function of S.

(b) Show that the Helmholtz free energy $F = E - TS$ is a concave function of T.

5. **Antiferromagnetism (50 points)**

Consider the antiferromagnet Ising model on a d-dimensional square lattice in a uniform magnetic field h:

$$H = +J \sum_{\langle ij \rangle} s_i s_j - h \sum_i s_i, \quad J > 0.$$

Take s_i to be Ising spins with values ± 1. There are two order parameters in this problem, the magnetization per spin $s = \langle s_i \rangle$ and the staggered magnetization per spin $t = \langle (-1)^i s_i \rangle$. Complete antiferromagnetic ordering occurs when $t = \pm 1$, $s = 0$; complete ferromagnetic order when $s = \pm 1$, $t = 0$.

(a) Find the ground state of this system at zero temperature as a function of the magnetic field strength h. Draw the 1D phase diagram (in h), indicating where the transitions occurs. *Hint: Show that at small $|h|$, the system display perfect antiferromagnetic order. However, if $|h| \geq h_c$, the system prefers to have perfect ferromagnetic order. Find h_c. Is the transition first or second order?*

(b) Analyze this system using mean field theory. Let $s_e = \langle s_i \rangle$ be the average magnetization per spin on even sites ($i = 2, 4, 6, \dots$) and $s_o = \langle s_i \rangle$ be the average magnetization per spin on odd sites ($i = 1, 3, 5, \dots$). Explain why we can write $s = (s_e + s_o)/2$ and $t = (s_e - s_o)/2$. The effective single-site Boltzmann factor for even site, s_i, $i \in$ even, is $e^{-2dJ\beta s_o s_i + \beta h s_i}$. Show that

$$\langle s_e \rangle = \tanh(2d\beta J(t - s) + \beta h).$$

Likewise, show that for, s_i, $i \in$ odd, we get

$$\langle s_o \rangle = \tanh(-2d\beta J(t + s) + \beta h).$$

Hence, show that the set of self-consistent equations involving the two order parameters s and t are

$$s = \frac{1}{2}\left[\tanh(2d\beta J(t - s) + \beta h) - \tanh(2d\beta J(t + s) - \beta h)\right], \tag{1}$$

$$t = \frac{1}{2}\left[\tanh(2d\beta J(t - s) + \beta h) + \tanh(2d\beta J(t + s) - \beta h)\right]. \tag{2}$$

(c) Analyze the mean field equations derived in part (b) at $h = 0$. Find the ground state of this system at $h = 0$ as a function of the temperature T. Indicate the critical temperature T_c. *Hint: consider the three cases: (1) $s = 0$, $t = 0$, (2) $s = 0$, $t \neq 0$, and (3) $t = 0$, $s \neq 0$ carefully.*

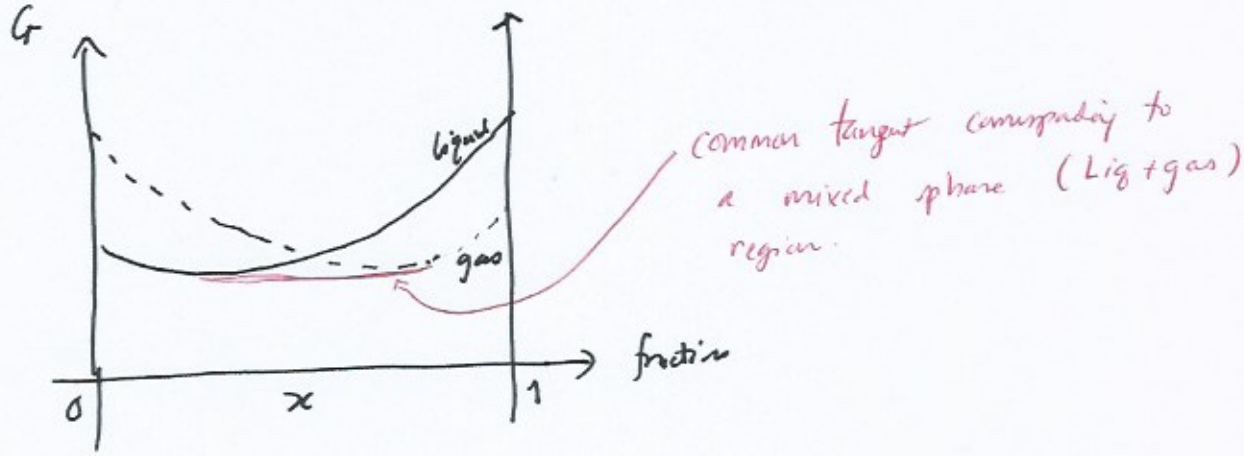
(1)

$$\langle x^2 \rangle = \frac{1}{a^2}$$

$$\langle x^6 \rangle = \langle x x x x x x \rangle + \langle x x x x x x \rangle + \langle x x x x x x \rangle + \cdots$$

$$= \frac{5 \cdot 3 \cdot 1}{a^6} = \frac{15}{a^6}$$

(2)

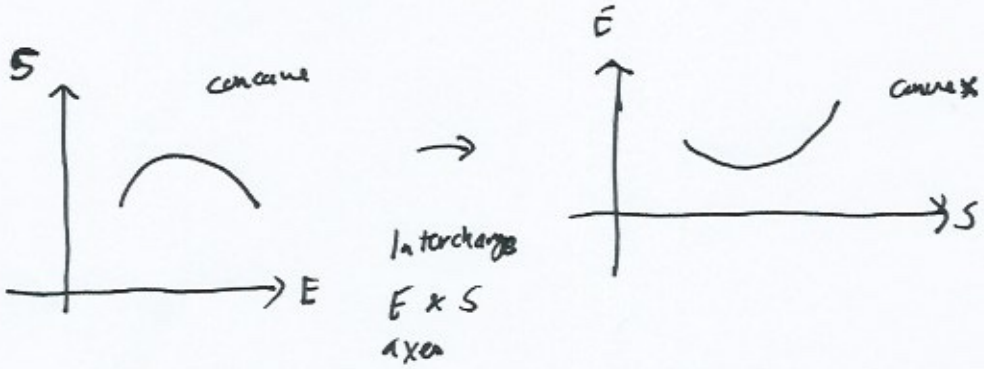


(3)

$$\lambda = \lim_{t \to 0} \frac{\ln |f(t)|}{\ln |t|} = \lim_{t \to 0} \frac{\frac{2}{5} \ln|t| + \frac{1}{4} \ln(t+B)}{\ln|t|} = 2/5$$

(4)

a)



b)

$$F = E - TS$$

E is convex function of S

$$dF = dE - TdS - SdT$$

1st law of Thermodynamics $\longrightarrow$ TdS

$$= -SdT$$

$$F = -k_B T \log Z$$

can solve using this method like done in lecture

$$\frac{\partial F}{\partial T} = -k_B \beta^2 \frac{\partial}{\partial \beta} \left(-\frac{1}{\beta} \ln Z \right)$$

$$\vdots$$

$$\frac{\partial^2 F}{\partial T^2} = -k_B^2 \beta^2 \langle (\delta E)^2 \rangle \sim -C_V$$

$$\leq 0 \quad \text{(concave)}$$

$$\therefore \quad \frac{\partial F}{\partial T} = -S$$

$$T = \frac{\partial E}{\partial S}$$

$$\Rightarrow \quad \frac{\partial^2 F}{\partial T^2} = -\frac{\partial S}{\partial T} = -\frac{1}{\frac{\partial T}{\partial S}}$$

$$= -\frac{1}{\frac{\partial^2 E}{\partial S^2}} \leq 0 \qquad \text{since } \frac{\partial^2 E}{\partial S^2} \geq 0 \quad \left(\begin{array}{c}\text{convex} \\ \text{function in } S\end{array}\right)$$

5)

Antiferromagnetism

(a) $T = 0$

Perfect antiferromagnetic order $\Rightarrow$ $\langle t \rangle = \pm 1$, $\langle S \rangle = 0$

energy for 1 spin $= +J \cdot d \cdot (-1) = -Jd$

$$\begin{array}{cccc} \uparrow & \downarrow & \uparrow & \downarrow \\ \downarrow & \uparrow & \downarrow & \uparrow \\ \uparrow & \downarrow & \uparrow & \downarrow \end{array}$$

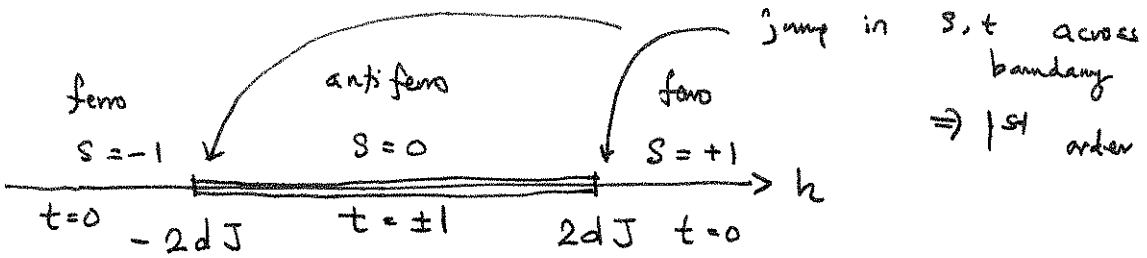
$$E_A = +J(-1)\, d \cdot N$$
$$= -JdN$$

complete ferromagnetic $\Rightarrow$ $\langle t \rangle = 0$, $\langle S \rangle = \pm 1$

$h > 0$ $\qquad\qquad$ $h < 0$

$$\begin{array}{cccc} \uparrow & \uparrow & \uparrow & \uparrow \\ \uparrow & \uparrow & \uparrow & \uparrow \end{array} \quad \text{or} \quad \begin{array}{cccc} \downarrow & \downarrow & \downarrow & \downarrow \\ \downarrow & \downarrow & \downarrow & \downarrow \end{array}$$

$$E_F = +J\, d\, N\, (+1) - hN$$
$$= JdN - Nh = N(dJ - h)$$



$$E_A = E_F \Rightarrow -JdN = JdN - Nh$$
$$\Rightarrow \boxed{h_c = 2dJ}$$

1st order phase transition

b) $s_e = \langle s_i \rangle$ = average spin on <u>even</u> sites

$s_o = \langle s_i \rangle$ = —— " —— <u>odd</u> sites

$$s = \text{average spin} = \frac{\text{average spin on even + odd sites}}{2}$$

$$= \frac{s_e + s_o}{2}$$

Similarly,

$$t = \frac{s_e - s_o}{2} \qquad \text{follows from. } t = \langle (-1)^i s_i \rangle$$

$$\Rightarrow \quad s_e = s + t$$

$$s_o = s - t$$

$$\langle s_e \rangle = \frac{(+1)\, e^{-2dJ\beta s_o(+1) + \beta h(+1)} + (-1)\, e^{-2dJ\beta s_o(-1) + \beta h(-1)}}{e^{-2dJ\beta s_o(+1) + \beta h(+1)} + e^{-2dJ\beta s_o(-1) + \beta h(-1)}}$$

$$= \tanh(-2dJ\beta s_o + \beta h)$$

$$= \tanh(2d\beta J(t - s) + \beta h)$$

Similarly the single-site Boltzmann factor for odd site is

$$e^{-2dJ\beta s_e s_i + \beta h s_i}$$

$$\Rightarrow \quad \langle s_0 \rangle = \tanh\left(-2dJ\beta\, s_e + \beta h\right)$$

$$= \tanh\left(-2dJ\beta(t+s) + \beta h\right)$$

Thus self consistency of the mean field solution means

$$s + t = \tanh\left(2d\beta J(t-s) + \beta h\right)$$

$$s - t = \tanh\left(-2d\beta J(t+s) + \beta h\right)$$

$$\therefore \quad s = \frac{1}{2}\left[\tanh\left(2d\beta J(t-s) + \beta h\right) - \tanh\left(2d\beta J(t+s) - \beta h\right)\right]$$

$$t = \frac{1}{2}\left[\tanh\left(2d\beta J(t-s) + \beta h\right) + \tanh\left(2d\beta J(t+s) - \beta h\right)\right]$$

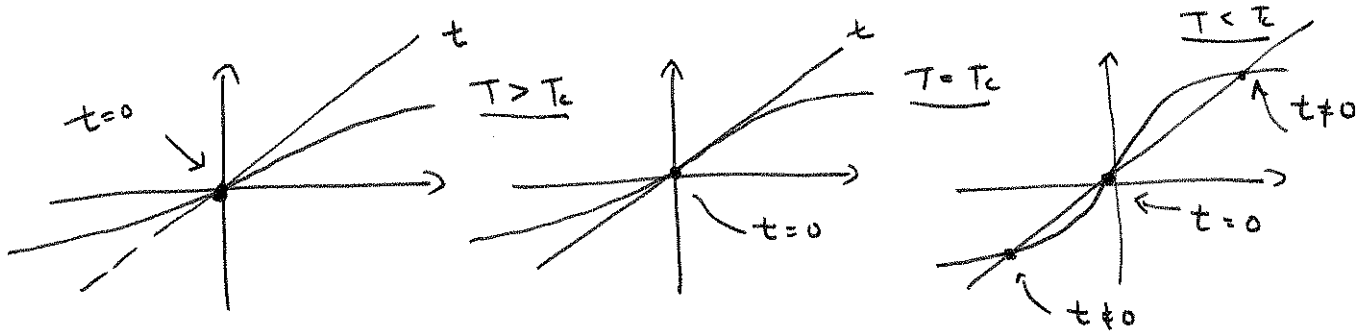
$\boxed{h=0}$

(c) $\underline{s = t = 0}$ is solution but a <u>trivial</u> one.

$\Rightarrow$ This is a <u>disorder</u> phase

$\underline{s = 0 \quad t \neq 0}$

$$\Rightarrow \quad t = \frac{1}{2}\left[\tanh(2d\beta J t) + \tanh(2d\beta J t)\right] = \tanh(2d\beta J t)$$



This is mean field equation of Ising model

$$t \simeq 2d\beta J t - \frac{(2d\beta J t)^3}{3} + \cdots$$

$$2d\beta_c J = 1$$

or $\boxed{T_c = 2dJ}$ <u>critical</u> <u>point</u>.

If t = non-trivial solution of the <u>Mean</u>-<u>field</u> equation

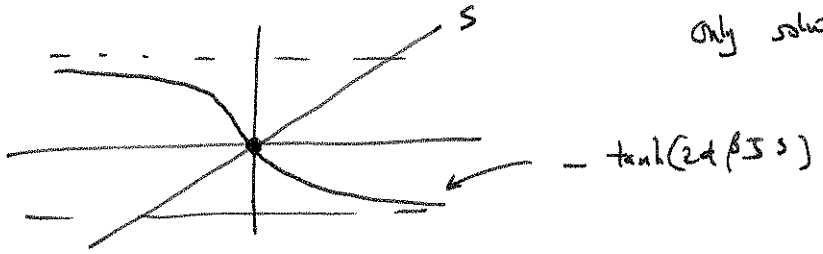
$s = 0$

that solves both equations.

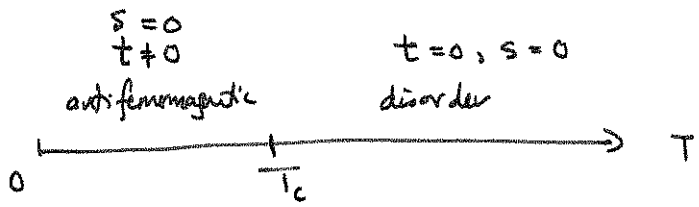
<u>$s \neq 0$, $t = 0$</u>

We find $s = \frac{1}{2}\left[\tanh 2d\beta J(-s) - \tanh 2d\beta J s\right]$

$$= -\tanh 2d\beta J s$$



Only solution is $s = 0$

which has <u>no</u> non-trivial self-consistent solution

Hence, we find